DATA PRIVACY CASE STUDY

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Title

Privacy Challenges in AI-based Medical Diagnostic Systems: A Federated Learning Approach for Secure Disease Prediction and Public Health Surveillance

Background

Healthcare organizations generate enormous volumes of patient information every day, including medical history, symptoms, diagnoses, laboratory reports, prescriptions, and treatment outcomes. AI systems require this data to identify disease patterns and improve clinical decision-making.

Traditional AI systems rely on centralized databases where patient information from multiple hospitals is collected in one location. Although this improves model performance, it introduces significant privacy and security risks.

Existing System

Traditional healthcare AI systems typically follow a centralized architecture.

Working

- Hospitals collect patient data.
- Patient records are uploaded to a centralized server.
- A machine learning model is trained using the combined dataset.
- Predictions are generated from the centralized model.

Limitations

- Raw patient data leaves hospitals.
- Large centralized databases become attractive targets for cyber attacks.
- High infrastructure costs.
- Privacy regulations restrict data sharing.
- Hospitals are reluctant to share confidential patient records.

- Single point of failure.

Proposed Solution

Proposed Federated Learning System

The proposed system follows a decentralized learning architecture.

Workflow

1. Each hospital stores patient records locally.
2. Local AI models are trained independently.
3. Only model parameters are transmitted.
4. Model updates are encrypted before transmission.
5. The central server aggregates updates from all hospitals.
6. An improved global AI model is generated.
7. The updated model is distributed back to all hospitals.
8. No raw patient information ever leaves the hospital.

Objectives

1) To analyse the privacy risks associated with AI-based medical diagnostic systems that rely on centralized collection and processing of sensitive patient data.
2) To study how Federated Learning can be used as a privacy-preserving machine learning approach that enables collaborative model training without sharing raw patient information.
3) To identify and evaluate privacy-preserving techniques such as encryption, differential privacy, secure aggregation, and access control in protecting healthcare data throughout the AI lifecycle.
4) To compare centralized AI architectures with decentralized Federated Learning architectures in terms of privacy, security, scalability, and regulatory compliance.
5) To examine the ethical, legal, and security implications of deploying AI-driven healthcare systems that process sensitive medical information.

Privacy-Preserving Technologies Used

The proposed system employs multiple privacy-preserving mechanisms to ensure secure collaborative learning.

Federated Learning

Allows hospitals to collaboratively train AI models without exchanging patient records.

Differential Privacy

Adds carefully controlled noise to model updates, reducing the possibility of reconstructing sensitive patient information from transmitted parameters.

Encryption

Protects model updates during transmission between hospitals and the central aggregation server.

Secure Aggregation

Ensures that the server aggregates encrypted model updates without learning individual hospital contributions.

Local Data Storage

Patient records remain within each hospital and are never transferred outside the institution.

Role-Based Access Control (RBAC)

Restricts access to sensitive patient information based on authorized user roles.

Audit Logging

Maintains records of data access and model updates to ensure accountability and traceability.

These mechanisms are part of the proposed architecture, which also includes encrypted local databases, audit logging, secure aggregation, and differential privacy during federated training.

Expected Outcomes

The proposed system is expected to

- Preserve patient privacy throughout the learning process.
- Eliminate the need for centralized patient databases.
- Improve disease prediction accuracy.
- Provide better treatment recommendations.
- Detect disease outbreaks at an earlier stage.
- Enable collaborative learning among hospitals.
- Reduce privacy risks associated with centralized healthcare AI systems.
- Improve clinical decision-making using collective knowledge.